

Effector V δ 1 $\gamma\delta$ T cells mediate anti-tumor immunity and are reprogrammed by imatinib in human gastrointestinal stromal tumor

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Effector V δ 1 $\gamma\delta$ T cells mediate anti-tumor immunity and are reprogrammed by imatinib in human gastrointestinal stromal tumor

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Abstract

Gamma-delta ($\gamma\delta$) T cells are emerging effectors of anti-tumor immunity, but their relevance to gastrointestinal stromal tumor, the most common human sarcoma, is not defined. Here, we integrate single-cell transcriptomics, immunophenotypic and T cell receptor profiling, and functional assays of circulating and intratumoral $\gamma\delta$ T cells across 68 human GIST specimens. V δ 1 cells predominated within tumors, and their abundance correlated with improved outcomes across 3 separate patient cohorts. $\gamma\delta$ T cells from untreated tumors were cytotoxic *ex vivo*, but tumors that had become resistant to the tyrosine kinase inhibitor imatinib had lower V δ 1 composition, effector function, and altered clonality, and were enriched for apoptosis, IL-17 signaling, and checkpoint pathways. In a genetically engineered murine GIST model, PD-L1 blockade enhanced imatinib efficacy and restored tumor $\gamma\delta$ T cell function. Thus, $\gamma\delta$ T cells contribute to tumor immunity in GIST and are reprogrammed during imatinib resistance, making them an attractive immunotherapy target.

Introduction

Gastrointestinal stromal tumor (GIST) is the most common human sarcoma and usually harbors either a *KIT* or *PDGFRA* mutation¹. The tyrosine kinase inhibitor (TKI) imatinib mesylate improves survival but is rarely curative due to acquired drug resistance, which typically occurs within 2 years¹. Although we found that imatinib initially increased intratumoral CD8⁺ T cells in a genetically engineered *Kit*^{V558del/+} mouse model of GIST^{2,3}, in human GIST *CD4* and *CD8A* expression were not prognostic⁴ and conventional T cell checkpoint blockade was ineffective⁵; these findings prompted us to examine other immune effectors.

$\gamma\delta$ T cells recognize a variety of antigens on tumor cells and unlike $\alpha\beta$ T cells, exert cytotoxicity independently of HLA⁶. There are different subsets of human $\gamma\delta$ T cells based on their γ and δ chains. Each TCR δ chain associates with one or more dominant TCR γ chains. The V γ 9V δ 2 subset predominates in human peripheral blood and can be readily expanded *in vivo* and *in vitro*. In contrast, human solid tissues are enriched for V δ 2-negative $\gamma\delta$ T cells, primarily V δ 1 cells, which perform immune surveillance and possess cytotoxic potential⁶. A smaller V δ 3 subset, most often paired with V γ 2, has been described, although its functional role is unclear. Effects of $\gamma\delta$ T cells in human solid tumors depend on tumor type and $\gamma\delta$ T cell subset composition⁷. Anti-tumor effects are driven through interferon- γ (IFN γ) production and innate-like cytotoxicity; increased $\gamma\delta$ T cell tumor infiltration has been associated with better clinical outcome in some malignancies^{8,9}. However, in other contexts, tumor-infiltrating $\gamma\delta$ T cells may acquire inhibitory phenotypes associated with impaired effector function and reduced IFN γ secretion¹⁰. IL17A-producing $\gamma\delta$ T cells have variable effects on tumors, as they are anti-tumoral in colorectal cancer¹¹ but mediate tumor resistance in breast cancer¹². In murine models, single-cell RNA sequencing and $\gamma\delta$ TCR profiling have demonstrated that $\gamma\delta$ cell function is strongly influenced by TCR usage¹³. Individual $\gamma\delta$ TCRs have an intrinsic ability to participate in both innate and adaptive immune features driven by CDR3 δ length and composition¹⁴.

Here, we establish a $\gamma\delta$ T cell atlas in human GIST by integrating single-cell transcriptomics, immunophenotyping, TCR profiling, and functional assays of circulating and tumor-infiltrating $\gamma\delta$ T cells across 68 tumor samples. Our findings establish $\gamma\delta$ T cells as a core component of GIST immunosurveillance and provide a framework for $\gamma\delta$ T cell-directed immunotherapy in GIST.

Results

Single-cell transcriptomics reveal $\gamma\delta$ T cell heterogeneity.

To dissect the role of $\gamma\delta$ T cells in human GIST, we sorted CD45⁺CD3⁺TCR $\gamma\delta$ ⁺ cells from untreated human GISTs and matched peripheral blood mononuclear cells (PBMCs) to profile them using single-cell RNA sequencing (scRNA-seq) (**Fig. S1, S2A-B**). After quality control, we obtained 55,228 cells from 4 blood and 2 tumor samples. We initially defined V δ 1, V δ 2, and V δ 3 T cell clusters based on canonical marker (*TRDV1*, *TRDV2*, and *TRDV3*) expression (**Fig. 1A**). V δ 2 cells were more prevalent in PBMCs, while V δ 1 and V δ 3 subsets were more frequent in tumors, consistent with reports that V δ 2-negative cells predominate within solid tumors^{15,16}. Flow cytometry in 13 patients revealed that the percentages of V δ 1 T cells correlated between PBMCs and tumors (**Fig. S3**).

Unbiased clustering of our integrated tumor/PBMC $\gamma\delta$ T cell atlas demonstrated 7 clusters for V δ 1, 4 for V δ 2, and 1 for V δ 3 based on differential gene expression (**Fig. 1B-C, S2C-D, Table S1**). The V δ 1-Cytotoxic T lymphocyte (CTL) cluster highly expressed classic markers for effector function (**Fig. 1D**), such as cytotoxicity (*GZMH*, *NKG7*, *KLRD1*, *KLRK1*, *GNLY*, *KLRF1*) and other effector and exhaustion markers (*FCGR3A*, *FCRL6*, *NCR1* (NKp46), *CD244*, *CD8A*, *KLRC2*, *IFNG*, and *TIGIT*). There was also high expression of *TBX21*, *ZEB2*, and *EOMES*, the master transcriptional regulators of conventional T cell effector function. In contrast, the V δ 1-naïve cluster possessed high expression of innate markers (*TCF7*, *LEF1*, *LTB*, *XCL1*, *CCR7*, *MAL*) as well as

the transcription factor *ZNF683* encoding *PRDM1* (BLIMP1) that regulates T cell differentiation and lineage. The V δ 1-Inflammatory cluster had the highest expression of *KLRK1* (NKG2D), *IL2RB*, and *IKZF2* (Helios) and increased expression of genes involved in T cell-mediated inflammation and activation (*AOAH*, *PRKCH*, *ZEB2*, *PIK3AP1*, *BCL2*, *RUNX1*). The V δ 3 subset most closely resembled the V δ 1-CTL cluster and had the highest expression of killer-cell immunoglobulin-like receptors (KIRs) and *GNLY*, implying cytotoxicity based on HLA class I expression.

Of the 4 V δ 2 T cell clusters, V δ 2-CTL predominated and was enriched for hallmark V δ 2 markers (*CEBPD*, *KLRB1*, *GZMK*, *KLRC1*, *IL7R*, and *CD23R*)¹⁷. Compared to V δ 1-CTL, V δ 2-CTL had lower expression of cytotoxic molecules, exhaustion markers, and *IFNG*. Another V δ 2 cluster highly expressed *CD44*, *CD69*, *ZBTB16* (encoding *PLZF*), *RORA*, *RUNX1*, *CAMK4*, a calmodulin-dependent protein kinase (CaMK), and *BTN3A1*, an essential membrane protein for TCR-dependent activation of human V γ 9V δ 2 T cells¹⁸. This V δ 2 cluster was also highly enriched for other unique markers (e.g., *ANXA1*, *IL17RA*, *IL23R*, *IFNGR1*, *CXCR6*, *AREG*, *THEMIS*) and was annotated as V δ 2-type 3 immunity, based on a prior report¹⁹. Smaller populations of V δ 2 T cells (V δ 2-Naïve and V δ 2-Differentiation) shared some common gene markers with V δ 2-CTL, but showed higher expression of *SELL* and *CCL4*, respectively.

Tumor $\gamma\delta$ T cells had unique gene expression programs compared to PBMC $\gamma\delta$ T cells (**Fig. 1E, left**), including heat shock proteins (*HSPA1B* and *HSPA1A*), stress response genes (*DNAJB1*), the integrin *ITGA1*, and other differentiation-related genes (e.g., *ZNF683*, *XCL1*, *DUSP4*, and *CXCR3*). Gene set enrichment analysis (GSEA) revealed multiple signaling pathways in tumor-infiltrating versus circulating $\gamma\delta$ T cells, such as cytokine-receptor interactions, inflammatory response, IL2/STAT5 signaling, and IFN γ (**Fig. 1E, right**). Compared to tumor V δ 2 cells, tumor V δ 1 cells upregulated most V δ 1 signature genes (e.g. *TRDV1*, *TRDV3*, *TRGV8*, *KLRD1*, *FCGR3A*, *KLRC2*, *KLRF1*, *IKZF2*, *FCRL6*, *XCL1*, *NCR1*, *ITGAD*) and other markers of

cytotoxicity and exhaustion. Similarly, tumor V δ 2 cells showed increased expression of V δ 2 signature genes (*TRDV2*, *TRGV9*, *IL7R*, *THEMIS*, *ANXA1*, *RORA*, and *RUNX2*; **Fig. 1F, top**). GSEA demonstrated that tumor V δ 1 cells were enriched for cell cycle pathways, oxidative phosphorylation, antigen processing, and NK cell-mediated cytotoxicity (**Fig. 1F, bottom**), the latter two of which were also enriched in circulating V δ 1 cells (**Fig. S2E**). Lastly, using CytoTRACE to infer relative differentiation states among $\gamma\delta$ T cell populations²⁰, tumor-infiltrating $\gamma\delta$ T cells were more differentiated than circulating $\gamma\delta$ T cells (**Fig. 1G**). Taken together, transcriptomes of purified PBMC and tumor $\gamma\delta$ T cells of GIST patients indicated considerable heterogeneity and broad functional plasticity.

Regulatory network analysis identifies V δ 1 $\gamma\delta$ T cell transcriptional programs.

To investigate the transcription factors driving $\gamma\delta$ T subsets, we used single-cell regulatory network interference and clustering (SCENIC) to identify potential regulatory mechanisms²¹. Regulons mediating lymphocyte trafficking (e.g., *RUNX3*, *KLF2*, *PRDM1*, *CEBPD*) were highly activated in circulating $\gamma\delta$ T cells across all clusters (**Fig. 2A-B**). Predicted *CEBPD* target genes included both V δ 1 and V δ 2 signature genes (e.g., *ZEB2*, *ZBTB16*, *ITGA1*, *THEMIS*, *RORA*, *IL7R*), which mediate T cell differentiation and activation (**Fig. 2C**). In addition, regulons of *EOMES*, *TBX21* (Tbet), and *RORC*, which regulate conventional T cell (Tconv) activation, were also higher in PBMC $\gamma\delta$ T cells (**Fig. 2A**). On further analysis, *RORC* was mainly detectable in V δ 2 (**Fig. 2B-C**), but not V δ 1 cells. Conversely, *EOMES* and *TBX21* were primarily in the V δ 1 subset, especially in the V δ 1-CTL cluster (**Fig. 2C**), suggesting they functionally resembled CD8⁺ $\alpha\beta$ T cells. Regulons linked to inflammation or T cell exhaustion induced by stress antigens (e.g., *STAT1*, *NFAT5*, *IRF8*) were enriched across all tumor-infiltrating $\gamma\delta$ T cells, suggesting presence of IFN γ signaling, inflammation, and exhausted T cell differentiation²²⁻²⁴.

Tumor V δ 1 and V δ 2 T cells had distinct regulon activity (**Fig. 2D, S4**). EOMES, TBX21, and POLR3G were enriched in V δ 1 cells, particularly in the V δ 1-CTL cluster (**Fig. 2C**). NFAT5 and IRF6 were specifically increased in the V δ 1-Inflammatory cluster. In contrast, RORC was more activated in the tumor V δ 2 subset. RORA and RUNX2 primarily regulated the V δ 2-Type 3 immunity phenotype (**Fig. 2C**). To identify potential functional differences mediated by these regulons, we computed immune module scores based on selected gene markers for memory, exhaustion, proliferation, activation, and cytotoxicity (**Table S2**)²⁵⁻²⁷. Tumor V δ 1 cells had higher exhaustion, activation, and cytotoxicity scores, whereas tumor V δ 2 cells had higher memory scores (**Fig. 2E**). Lastly, enrichment analysis using Gene Ontology was conducted based on predicted target genes of each regulon (**Fig. 2F**). Consistent with their transcriptional profiling, targets of EOMES and POLR3G were enriched for T cell activation, cytotoxicity, and lymphocyte chemotaxis. Meanwhile, RORC and RUNX2 were associated with pathways in T cell differentiation, activation, and leukocyte adhesion. In summary, we defined diverse functional phenotypes in $\gamma\delta$ T cells and inferred their regulatory mechanisms.

$\gamma\delta$ TCR analysis uncovers clonotype diversity and TKI alteration.

To examine the $\gamma\delta$ TCR repertoire, we performed next-generation sequencing of genomic DNA for TCR delta chains in sorted $\gamma\delta$ T cells from 5 pair-matched PBMC and tumor samples (3 untreated and 2 with acquired resistance to imatinib) and 1 additional resistant tumor. Most (>75%) of the PBMC and tumor $\gamma\delta$ TCRs were minimally expanded with fewer than 5 clones (**Fig. 3A**). Patients with resistant tumors had a higher frequency of highly expanded clones (clone size >10), particularly in the blood (**Fig. 3B**).

In 4 of 5 patients, TCRDV1 usage was higher in tumors than paired PBMCs (**Fig. 3C**), consistent with our scRNA-seq data. Notably, TCRDV1 usage was lower in the 2 tumors that developed resistance to imatinib compared to the 3 untreated tumors, while TCRDV2 gene usage

was higher in the 2 resistant tumors. The tree plot summarizes TCRDV usage. TCRDV1 usage was higher in tumors than PBMCs in the untreated samples, both overall and in highly expanded clones. Furthermore, it was higher in untreated versus resistant patients in the tumor compartment by both metrics (**Fig. 3D**). We next performed differential expansion analysis for each clone between paired PBMC and tumor samples (**Fig. 3E**). In resistant tumors, there was a higher frequency of highly expanded clones compared to untreated tumors (2.78 vs. 0.89%, $p=3.91\times 10^{-11}$). The recombinations V1D1J1 and V2D1J1 predominated in both groups, suggesting that the primary variation lies in V gene usage (**Fig. 3F**). In the untreated group, V1D1J1 and V2D1J1 appeared at a similar frequency, whereas in the resistant group, V2D1J1 was enriched while V1D1J1 was reduced (**Fig. 3G**). To confirm our findings, we performed flow cytometry for V δ 1 and V δ 2 in human PBMCs and tumors in 29 patients. Consistent with our scRNA-seq and TCR sequencing in sorted $\gamma\delta$ T cells, the V δ 1 subset dominated in the tumor, whereas the V δ 2 subset was more abundant in paired PBMCs (**Fig. 3H**). Furthermore, TKI-resistant GISTs had lower tumor V δ 1 and higher V δ 2 composition than untreated tumors (**Fig. 3I, left**), even though the relative proportion of the 2 subsets did not vary with treatment status in PBMCs (**Fig. 3I, right**). Overall, resistant tumors had an altered $\gamma\delta$ TCR repertoire.

Tumor V δ 1 T cells are cytotoxic and prognostic of clinical outcome.

To validate our scRNA-seq atlas, which showed effector and exhaustion markers in the V δ 1 subset, we performed flow cytometry on 10 untreated GISTs. Tumor-infiltrating V δ 1 T cells expressed CD8A, FCGR3A (encoding CD16), PDCD1 (PD1), TIGIT, and EOMES (**Fig. S5A**), as identified in Fig. 1D. In addition, expression of CCR7, HLA-DR, and the cytotoxic degranulation marker CD107a was higher in tumor-infiltrating V δ 1 cells compared to V δ 2 cells (**Fig. 4A, S5B**) and PBMC V δ 1 cells (**Fig. 4B**), indicating enhanced tumor trafficking, activation, and cytotoxic potential. To assess the functional relevance of tumor-infiltrating V δ 1 cells *ex vivo*, we sorted $\gamma\delta$

T cells from 9 additional human GISTs (7 untreated and 2 imatinib-resistant). Six tumors had a high proportion of V δ 1 cells among $\gamma\delta$ T cells (mean 59.2%), while 3 had V δ 1 lower percentages (mean 32.6%) (**Fig. 4C**). Tumors with a high V δ 1 proportion had increased expression of *TRGC2*, *TRDV3*, *IFNG* and *KLRK1* (encoding NKG2D) (**Fig. 4D**), thereby validating our scRNA-seq findings shown in Figure 1D.

The TCR data of sorted $\gamma\delta$ T cells suggested that resistant GISTs had altered clonotype diversity and subset composition. Therefore, we performed additional scRNA-seq analysis on 3 untreated and 2 resistant GISTs, without sorting $\gamma\delta$ T cells. CD3⁺ T cells were identified and clustered across all samples (**Fig. S6A**). Two clusters were annotated as $\gamma\delta$ T cells based on canonical gene expression (**Fig. S6B**). Using reference mapping from our $\gamma\delta$ T cell scRNA-seq atlas (**Fig. 1A**), we identified a substantial shift in $\gamma\delta$ T cell subset composition, from V δ 1 dominance in untreated tumors to a more balanced V δ 2 proportion in resistant tumors (**Fig. 4E**). Differential gene expression analysis revealed enrichment of tumor-specific V δ 1 signature genes in untreated tumor $\gamma\delta$ T cells and V δ 2 genes in resistant $\gamma\delta$ T cells (**Fig. 1F, 4F, left**). Notably, higher expression of cytotoxic effector genes, including *GNLY*, *PRF1*, *KLRK1*, *KLRD1*, and *FCGR3A*, was observed in untreated tumors, indicating enhanced cytotoxic activity (**Fig. 4F, right**). Over-representation enrichment analysis (ORA) confirmed that untreated tumor $\gamma\delta$ T cells were significantly enriched for immune effector processes, leukocyte-mediated immunity, cell killing, and NK cell-mediated cytotoxicity. In contrast, $\gamma\delta$ T cells in resistant tumors had increased apoptosis, T cell receptor signaling, and lipid metabolism, as well as PD-1/PD-L1 checkpoint, NF-kappa B, and IL-17 signaling pathways (**Fig. 4G**).

To further validate these findings, we analyzed RNA from 29 human GISTs (15 untreated and 14 resistant tumors) by quantitative real-time PCR (qPCR). Expression of *TRGC2* and *TRDV3*, reflecting the abundance of V δ 1 and V δ 3 cells, respectively, trended lower in imatinib-resistant tumors than in untreated GISTs (Fig. 4H), corroborating our flow cytometry and scRNA-

seq data showing reduced V δ 1 cells in resistant GISTs, with concordant IFNG reduction. CD107a expression was decreased in tumor-infiltrating V δ 1 cells from resistant GISTs (**Fig. 4I**), indicating reduced effector function. V δ 1 T cells in untreated tumors predominantly displayed an effector phenotype, with high frequencies of T cell effector memory (T_{EM}) populations, in contrast to resistant tumors (**Fig. 4J**).

To determine the clinical relevance of tumor V δ 1 T cells in GIST, we analyzed 3 independent cohorts of untreated patients. In a bulk tumor RNA microarray of 32 untreated primary human GISTs from Japan²⁸, we applied a published $\gamma\delta$ T cell signature (*TRGC2*, *TRDC*, *TRDV3*, and *TRGV2*)²⁹ and stratified high versus low signature expression based on the median. High signature expression was associated with improved overall survival (**Fig. 4K**). Next, we evaluated this $\gamma\delta$ T cell signature in our published cohort of 30 untreated primary GISTs³⁰ and found that it correlated with predicted recurrence-free survival (RFS)³¹ (**Fig. 4L**). Finally, we quantified V δ 1 T cell abundance by flow cytometry in 17 untreated primary GIST tumors and again found a correlation to predicted RFS (**Fig. 4M**). Thus, $\gamma\delta$ T cells, and V δ 1 T cells in particular, are favorably prognostic in untreated human GIST.

$\gamma\delta$ T cells are cytotoxic in patient-derived tumor models.

To assess whether $\gamma\delta$ T cells kill GIST tumors, we first expanded PBMC-derived $\gamma\delta$ T cells from healthy donors (n=4) with an anti- $\gamma\delta$ TCR antibody and IL-2 (**Fig. S7A**)³². Healthy donors with a high frequency of V δ 1 cells after 8 days of expansion (**Fig. S7B**) were used for *in vitro* assays (n=2). Expanded $\gamma\delta$ T cells demonstrated dose-dependent cytotoxicity against luciferase-expressing human GIST T1 cells (**Fig. 5A**). Fluorescently labeled $\gamma\delta$ T cells migrated into GIST spheroids and accomplished tumor cell killing, albeit at a higher E:T ratio (**Fig. 5B-C**). To evaluate $\gamma\delta$ T cell function in a more physiologic setting, we utilized an *ex vivo* human GIST tumor slice model. Primary tumor slices from 3 untreated patients were co-cultured with expanded $\gamma\delta$ T cells

from healthy donors. After 24h, tissue slices were dissociated into single-cell suspensions and viable KIT⁺ tumor cells were quantified by flow cytometry. Expanded $\gamma\delta$ T cells infiltrated tumor slices (**Fig. S7C**) and reduced KIT⁺ tumor cells (**Fig. 5D**).

Although tumor-infiltrating $\gamma\delta$ T cells are thought to possess limited anti-tumor activity in certain cancers due to their exhausted phenotypes⁶, our scRNA-seq data revealed enrichment of a NK cell-mediated cytotoxicity pathway in tumor V δ 1 cells. Therefore, we expanded tumor-infiltrating lymphocytes (TILs) from 2 untreated GIST patients for 6 days using an anti- $\gamma\delta$ TCR antibody and IL-2, and cultured them with matched tumor single-cell suspensions for 24h (**Fig. 5E-F**). Co-culture reduced the number of viable KIT⁺ tumor cells (**Fig. 5G**). Given that tumor-associated ligands can activate $\gamma\delta$ T cells, we examined the role of MICA, an NKG2D ligand expressed by KIT⁺ tumor cells in human GIST (**Fig. 5H**). Plate-bound recombinant MICA plus IL-2 expanded V δ 1 cells within TILs and induced higher expression of HLA-DR (**Fig. 5I**). MICA stimulation also induced IFN γ production, which was attenuated by NKG2D blockade (**Fig. 5J**). Thus, both circulating and tumor-infiltrating $\gamma\delta$ T cells, particularly the V δ 1 subset, retain cytotoxic effector function following expansion and activation.

Initial imatinib response alters tumor $\gamma\delta$ T cell composition and function in mice.

Our human studies utilized untreated tumors and tumors that acquired resistance to imatinib. To investigate the early effect of imatinib on tumor $\gamma\delta$ T cells, we administered imatinib for 1 week to *Kit*^{V558del/+} mice³, which develop an intestinal GIST. Bulk RNA-seq of sorted tumor $\gamma\delta$ T cells showed that imatinib upregulated 414 genes and downregulated 577 (**Fig. 6A**). GSEA revealed increased oxidative phosphorylation, fatty acid metabolism, and glycolysis/gluconeogenesis pathways. Murine IL17A-secreting $\gamma\delta$ T cells rely on oxidative lipid metabolism³³, and previously we found that imatinib expanded mouse IL17A-producing $\gamma\delta$ T cells⁴. Here, we further identified that imatinib reduced $\gamma\delta$ T cell IFN γ response, antigen presentation, and NK cell-mediated

cytotoxicity (**Fig. 6B-C**). IFN γ -producing murine $\gamma\delta$ T cells, particularly the V γ 7/V γ 1 subset, display antitumor activity in various models, whereas IL-17A-producing $\gamma\delta$ T cells, mainly the long-lived V γ 6 subsets, have variable pro-tumoral function¹³. In our model, by bulk gene expression and our published murine scRNA-seq data⁴, V γ 6 cells (cluster 1) predominated and were expanded by imatinib, while the V γ 7/V γ 1 subset (cluster 4) was reduced (**Fig. 6D-F, S8**). Accordingly, imatinib upregulated $\gamma\delta$ -Th17 signature genes and reduced $\gamma\delta$ -Th1 related and cytotoxicity genes (e.g., *Gzmb*, *Klrd1*, *Klrc2*), respectively. Bulk RNA-seq also showed that imatinib increased expression of *Pdcd1* (PD1) and *Tnfrsf9* (4-1BB ligand receptor), which were validated by qPCR (**Fig. 6G**).

Since checkpoint inhibition of PD-1 and PD-L1 can augment $\gamma\delta$ T cell cytotoxicity in other solid tumors^{16,34}, we hypothesized that targeting the PD1/PDL1 pathway would affect $\gamma\delta$ T cell immunity in GIST. The combination of imatinib and an anti-PD-L1 blocking antibody further reduced tumor weight as expected³⁵, but also restored the proportion of V γ 7⁺ cells, downregulated their PD1 expression, and increased $\gamma\delta$ T cell IFN γ while maintaining IL-17 production (**Fig. 6H-L**). Taken together, these data suggest that checkpoint inhibition increased the anti-tumor efficacy of $\gamma\delta$ T cells during imatinib response.

Discussion

By integrating single-cell transcriptomic, TCR repertoire profiling, and functional analyses, we established a comprehensive framework for understanding tumor and circulating $\gamma\delta$ T cell heterogeneity, differentiation, clonotype diversity, and function in GIST, revealing previously unappreciated mechanisms linking oncogenic signaling, innate-like cytotoxic immunity, and therapeutic response. V δ 1 cells were more prevalent in human tumor samples than in matched blood, regulated by EOMES and TBX21, and enriched for a T_{EM} phenotype. V δ 1 cells expanded *in vitro* from healthy donor PBMCs killed tumor cells in 3 different patient-derived models, including freshly procured GIST tumor slices. TILs from GIST specimens expanded with an anti-TCR antibody were also cytotoxic against autologous tumor cells. The baseline $\gamma\delta$ T frequency in PBMCs influence *ex vivo* expansion capacity³⁶. In contrast, V δ 2 cells were driven by RORC, predominated in tumors that had acquired resistance to imatinib, and expressed fewer effector molecules. Analysis of 3 independent patient cohorts demonstrated the prognostic value of total, as well as V δ 1-specific, $\gamma\delta$ T cell abundance in untreated GISTs, as observed in other tumors.²⁹ Notably, including V δ 1- and V δ 3-specific genes (*TRGC2* and *TRDV3*, respectively) improved upon the prognostication of *TRDC* alone, which we used previously⁴.

The TCR repertoire of sorted $\gamma\delta$ T cells provides insight into the mechanisms underlying $\gamma\delta$ T cell activation in GIST. Untreated GIST tumors displayed broad V δ 1 TCR diversity with relatively few highly expanded clones, implying limited engagement of a few dominant immunogenic tumor antigens. Known $\gamma\delta$ TCR ligands, including Ephrin type-A receptor 2 (EphA2) and HLA/MHC class I chain-related proteins A and B (MICA/B), can trigger $\gamma\delta$ T cell differentiation and expansion within tumors^{37,38}. Indeed, MICA expression by human GIST tumor cells expanded the V δ 1 subset more efficiently than the V δ 2 subset and induced IFN γ production and HLA-DR upregulation. In contrast, the V δ 2 subset predominantly recognizes phosphoantigens (e.g., zoledronate) via the BTN2A1-BTN3A1 complex for expansion³⁹. Admittedly, our TCR analysis

focused on the δ chain, potentially underestimating clonotypic complexity. Nevertheless, these observations support the model that GIST contains multiple $\gamma\delta$ immune circuits, with V δ 1 cells preferentially linked to tumor stress recognition and direct cytotoxic effector function.

Imatinib therapy remodeled the $\gamma\delta$ T cell compartment in GISTs. While the V δ 1 subset was prevalent in untreated human GISTs, tumors that had acquired resistance to imatinib exhibited reduced V δ 1 representation, diminished T_{EM} differentiation, increased apoptosis, reduced cytotoxicity, altered cellular metabolism, and increased IL-17 signaling. These changes reinforce that while imatinib directly targets KIT signaling in tumor cells, the GIST tumor microenvironment is also remodeled, as we have shown previously with alterations in intratumoral T cells, tumor-associated macrophages, and dendritic cells^{2,40,41}.

Our murine *Kit*^{V558del/+} model allowed us to investigate $\gamma\delta$ T cells during the early (i.e., 1 week) tumor response to imatinib. Previously, we showed that $\gamma\delta$ T cells in GIST contribute to tumor control through interleukin-17 (IL-17) production, as antibody-mediated depletion of either IL-17 or $\gamma\delta$ T cells accelerated tumor growth in otherwise untreated mice⁴. The role of IL-17 in human GIST remains uncertain. Here, we found that the early response to imatinib in *Kit*^{V558del/+} mice upregulated tumor $\gamma\delta$ T cell fatty acid metabolism and oxidative phosphorylation, as detected by bulk RNA-seq analysis of sorted $\gamma\delta$ T cells. Furthermore, imatinib decreased tumor $\gamma\delta$ T cell IFN γ in addition to increasing IL-17 production. Combining PD-L1 blockade with imatinib increased tumor cytotoxicity, tumor V γ 7 cells, and tumor $\gamma\delta$ T cell IFN γ production. These findings are consistent with V δ 1 cells responding favorably to checkpoint inhibition in melanoma¹⁶ and Merkel cell carcinoma³⁴. Human GIST tumor samples during early response to imatinib are not clinically available. Moreover, technical challenges precluded the isolation and analysis of sufficient $\gamma\delta$ T cells after chronic therapy with imatinib, which is typically administered for at least 6 months prior to surgery to reduce tumor size.

339 Our study establishes V δ 1 $\gamma\delta$ T cells as critical mediators of anti-tumor immunity in GIST
340 and highlights $\gamma\delta$ T cells as an actionable immunotherapeutic axis in GIST and other solid tumors
341 that are refractory to conventional $\alpha\beta$ T cell-directed immunotherapy. Emerging clinical efforts to
342 harness $\gamma\delta$ T cells, including CAR-engineered V δ 1 therapies currently under investigation^{42,43},
343 further highlight the translational relevance of these immune cells.
344

Methods

Human GIST samples. Tumor specimens and PBMCs were obtained from patients with GIST undergoing surgical resection under an Institutional Review Board-approved protocol (additional supporting data files). All patients provided written informed consent prior to surgery, and studies were conducted in accordance with the Declaration of Helsinki. Patient information was de-identified and encrypted in compliance with the Health Insurance Portability and Accountability Act regulations (HIPAA). Tumor tissues were sectioned and processed into single-cell suspensions. PBMCs were isolated from whole blood using CPT tubes and density gradient centrifugation.

Cell isolation, sorting, and flow cytometry. Single-cell suspensions were prepared from PBMCs, human GISTs, and murine GISTs derived from *Kit*^{V558del/+} mice. In brief, tumor tissues were minced and digested with a mixture of 2.5 mg/ml collagenase IV and 50 μ g/ml DNase I at 37°C for 30 minutes. They were processed into single-cell suspensions as previously described³⁵. Flow cytometry was performed using an LSRFortessa (BD Biosciences). Prior to antibody staining, cells were incubated with an anti-CD16/32 antibody to block Fc receptors. The following fluorochrome-conjugated antibodies were used for murine cells: CD45 (30-F11), CD3 (17A2), CD4 (RM4-5), CD8 (53-6.7), NK1.1 (PK136), TCR $\gamma\delta$ (GL3), IL17A (TC11-18H10.1), PD-1 (29F.1A12), IFN- γ (XMG1.2), TCR V γ 1 (2.11), V γ 2 (UC3-10A6), V γ 4 (49.2), V γ 6 (1C10-1F7), V γ 7 (F2.67). For human samples, the following antibodies were used: CD45 (2D1), CD3 (SK7), CD8 (RPA-TB), TCR $\gamma\delta$ (B1), TCR V δ 1 (TS8.2), TCR V δ 2 (B6), TCR V γ 9 (B3), TIGIT (TgMab-2), CD45RA (HI100), PD-1 (EH12.2H7), CD69 (FN50), IFN- γ (4S.B3), CCR7 (G043H7), HLA-DR (G46-6), CD27 (L128), CD107a (H4A3), NKG2D (1D11), MICA/B (6D4). Viable cells were identified using LIVE/DEAD Fixable Aqua (Thermo Fisher Scientific). Transcription factor staining was performed with the Foxp3/Transcription Factor Staining Buffer Set (eBioscience), and

intracellular cytokine staining was conducted using the Cytofix/Cytoperm Kit (BD Biosciences). For cell sorting, fluorescent-labeled antibodies were used to stain single-cell suspensions, and sorting was performed on a FACS Aria III (BD Biosciences). Human $\gamma\delta$ T cells were defined as CD45⁺CD3⁺TCR $\gamma\delta$ ⁺, and murine $\gamma\delta$ T cells as CD45⁺CD3⁺NK1.1⁻TCR $\gamma\delta$ ⁺. Flow cytometry data were analyzed using FlowJo software (version 10.8.2; BD Biosciences).

Human $\gamma\delta$ T cell expansion and in vitro assays. PBMCs were isolated from the peripheral blood of healthy donors or GIST patients. Tumor infiltrating lymphocytes (TILs) were isolated from tumor single-cell suspensions using CD45 microbeads (Miltenyi Biotec). For $\gamma\delta$ T cell expansion, PBMCs or TILs (2x10⁶ cells per well) were seeded in 24-well plates pre-coated with anti- $\gamma\delta$ TCR antibody (0.5 μ g/ml; Immunotech) and cultured in RPMI1640 medium supplemented with IL-2 (200U/ml, PeproTech) for 6 to 10 days.

For *in vitro* cytotoxicity assays, luciferase-expressing GIST T1 cells (1x10⁴ per well) were seeded as target cells in 96-well flat-bottom plates. Expanded $\gamma\delta$ T cells were added as effector cells at the indicated effector-to-target (E:T) ratio and co-cultured for 24h. Cells were then lysed, and cytotoxicity was quantified using the Luciferase Assay System (E1501, Promega) on a GloMax Discover Microplate Reader, according to manufacturer's instructions. For three-dimensional (3D) spheroid assays, 96-well plates were pre-coated with 50 μ l of 1.0% agarose (Invitrogen). Luciferase-expressing GIST-T1 cells (5x10³ per well) were seeded and allowed to form spheroids for 24h. CellTrace-labeled $\gamma\delta$ T cells (CellMask deep red, Invitrogen) were then added at the indicated E:T ratios. $\gamma\delta$ T cell infiltration was analyzed by fluorescence microscopy at 24h and cytotoxicity was measured at 72h using the luciferase assay.

For GIST tissue slice co-culture assays, 6 mm tumor cores were collected in the operating room and maintained on ice. Specimens were embedded in low melting point agarose and sectioned into 300- μ m slices using a vibrating microtome as described previously⁴⁴. Tissue slices

were placed into 0.4 μ m Millicell inserts (Milipore) in 24-well plates and cultured in RPMI1640 medium supplemented with IL-2. One slice was dissociated to determine the number of target cells. CellTrace-labeled $\gamma\delta$ T cells were added to the top of tissue slices at an E:T ratio 5:1. After 24h of co-culture, slices were washed of free cells with PBS and dissociated into single-cell suspensions with collagenase and DNase I for flow cytometry analysis. To assess cytotoxicity of tumor-derived $\gamma\delta$ T cells against autologous KIT⁺ tumor cells, TILs were isolated and expanded for 6 days with an anti- $\gamma\delta$ TCR antibody in the presence of IL2. Expanded TILs were then co-cultured with autologous tumor single-cell suspensions at an E:T ratio of 5:1 for 24h *in vitro*. Live KIT⁺ tumor cells were quantified by flow cytometry and normalized to tumor single-cell suspensions cultured alone as the control.

For human MICA stimulation, TILs were cultured with recombinant MICA (rhMICA; 10 μ g/ml; R&D Systems) or mouse IgG1 isotype control (50 ng/ml; BioLegend) for 72h and analyzed by flow cytometry. To examine the role of NKG2D signaling, sorted $\gamma\delta$ T cells (predominantly V δ 1) from untreated GISTs were cultured in 96-well plates with rhMICA in the presence or absence of an anti-NKG2D blocking antibody (10 μ g/ml; clone 1D11; BioLegend) added prior to plating. IL-2 was included in the culture medium. After 24h, IFN γ production was measured by ELISpot (ImmunoSpot) following the manufacturer's instructions.

RNA isolation and real-time PCR. Total RNA was extracted from human GIST specimens and murine *Kit*^{V558del/+} tumors using the RNeasy Mini Kit (Qiagen). Reverse transcription was performed with TaqMan reagents (Applied Biosystems). TaqMan gene expression assays (Applied Biosystems) included the following probes: mouse *Pdcd1* (Mm01285676_m1), *Havcr1* (Mm01294183_m1) and *Tnfrsf9* (Mm00441899_m1); and human *Trgc2* (Hs00827007_m1), *Trdv3* (Hs01669899_m1), *Ifng* (Hs009899291_m1), *Mica* (Hs00792195_m1), and *Klrk1* (Hs00183683_m1). qPCR was performed using a QuantStudio real-time PCR system (Applied

Biosystems). Relative gene expression was calculated using the $2^{-\Delta\Delta CT}$ method, as described in the manufacturer's instructions, and results were expressed as fold increase for murine samples or as median values for human samples relative to the indicated controls.

Processing and quality control filtering of human $\gamma\delta$ T cell scRNA-seq data. Human $\gamma\delta$ T cells sorted from PBMCs and tumors were immediately processed for scRNA-seq using the 10x Genomics platform at the Center for Applied Genomics of the Children's Hospital of Philadelphia. Read count matrices from scRNA-seq data were generated from raw FASTQ files using CellRanger v.7.1.0, with alignment to the GRCh38-2020-A human transcriptome reference. The resulting count matrices were processed and analyzed using Seurat v5⁴⁵. Quality control filtering was applied to each cell, using filters of $100 < \text{nFeatureRNA} < 10000$ mitochondrial read percentage $< 10\%$. Doublets were then called using DoubletFinder (v2.0.4) using SCTransform normalization, an expected doublet rate of the total cell number / 125,000, and other parameters as default. Then, predicted doublets were removed and additional filters of $\text{nFeature_RNA} > 300$ and $\text{nCount_RNA} > 1000$ were applied.

Sample integration, clustering, and cell type annotation of human $\gamma\delta$ T cell scRNA-seq data. Analysis of scRNA-seq data was conducted using Seurat v5⁴⁵ and R v4.4.0. Samples were normalized individually using the SCTransform function with regression of the mitochondrial read percentage, and a PCA was computed. Then, samples were integrated using the IntegrateLayers function with RPCA integration and default parameters. A UMAP was constructed using the first 30 dimensions of the integrated reduction. Cells were clustered using the FindNeighbors function with the first 30 dimensions of the integrated reduction followed by the FindClusters function. The clustering resolution was chosen empirically. Cell cycle phase was predicted using the CellCycleScoring function after conversion of the default human gene lists to mouse orthologues.

Reference mapping was performed with the FindTransferAnchors and Mapquery functions. Differential gene expression was conducted using the FindMarkers function with the default Wilcoxon rank-sum test on log-normalized counts, and clusters were manually annotated based on their differentially expressed genes. For construction of volcano plots, cells were downsampled to a maximum of 1,500 per condition to avoid p-values of zero. For pathway analysis, the log2FC across all genes was used as input to a pre-ranked gene set enrichment analysis (GSEA) using the fgsea package. Pathways were sourced from the Molecular Signatures Database (MsigDB) Hallmark gene sets²⁷ and the KEGG database⁴⁶. Overrepresentation enrichment analysis was performed on the top genes with predefined filtering parameters from select gene set collections using the gost function from the gprofiler2 package (minimum expressing cells = 20%, |log-fc| > 1, FDR < 0.05). Differentiation trajectory was inferred using CytoTRACE (v0.3.3) with default parameters using the log-normalized counts.

SCENIC transcriptional regulatory network (TRN) analysis. The Single-Cell Regulatory Network Inference and Clustering (SCENIC) algorithm⁴⁷ was used to identify transcription factors regulating $\gamma\delta$ T cell states. It was run using the pySCENIC implementation as previously described⁴⁸ on the scRNA-seq data, after conversion of the Seurat object to loom format. The GRNBoost2 algorithm was used for GRN inference. To predict transcription factor regulons, we used the human v9 cisTarget motif collection and the hg38_refseq-r80 databases with the 500bpUp100Dw and TSS+/-10kb search spaces. All relevant databases were obtained from: <https://resources.aertslab.org/cistarget/>. The pySCENIC implementation of AUCell was used to score the activity of regulons for each cell. The SCoPeLoomR package (<https://github.com/aertslab/SCoPeLoomR>) was then used to extract the regulons and AUCell matrix from the resulting loom file, and the final AUC matrix was added to the initial Seurat object for visualization and analysis. Differential regulons were identified using the FindAllMarkers function using the Wilcoxon rank-sum test. Gene sets enriched within regulon target genes were

analyzed using the enrichGO function of clusterProfiler with the Biological Process Gene Ontology pathways.

TCR Sequencing and analysis of TCR α δ repertoires. Human $\gamma\delta$ T cells were sorted from PBMCs (n=5) and tumor samples (n=6), and genomic DNA (gDNA) was extracted. Next-generation DNA sequencing of the CDR3 regions of human TCR α and TCR δ chains was performed using the ImmunoSEQ® Assay (Adaptive Biotechnologies). gDNA was amplified by a multiplex PCR method, followed by high-throughput sequencing. Sequence output was filtered on in-frame CDR3s and normalized across all samples. All analyses were carried out using CDR3 amino sequences in the TCR δ chain to track clonal expansion within an individual. Briefly, to identify the TCR δ repertoires, data were pre-processed based on the productive rearranged TCR δ for each sample and TCRs with TCR α expression were first filtered out. Then, the count of TCR was quantified on the amino acid level for each sample. To measure clonotype frequency of TCR δ repertoires, we classified TCR into 5 expansion groups, using 1, 5, 10, 100 as boundaries according to the total count of TCR δ sequence in each sample. The frequency of each expansion group was calculated by the count of unique TCRs in the group divide by the total number of unique TCRs within the same sample. The frequency of the highly expanded group (clone size > 10) was calculated in the same way and compared between PBMC and tumor sample for each patient by the Wilcoxon rank-sum test. We applied Fisher's exact test to assess clonal expansion between paired PBMC and tumor samples for each individual clone. To compare the number of clones significantly expanded in the tumor relative to PBMC between untreated and resistant patients, we also used Fisher's exact test. Additionally, differences in the number of clones within each VDJ recombination group between the 2 patient groups were evaluated using the same test.

Mice and treatments. *Kit*^{V558del/+} mice (8-10 weeks old) on a B6 background (Jackson Laboratory) were maintained in a pathogen-free animal facility. Mice were age- and sex-matched and randomized for each experiment. Animal experiments were approved by the Institutional Animal Care and Use Committee at the University of Pennsylvania (IACUC). Imatinib mesylate (Novartis) was dissolved in the drinking water at 600 mg/L; isotype (rat IgG2b; BioXcell) and anti-PDL1 blocking antibody (10F.9G2; BioXcell) were administered intraperitoneally at a dose of 200 µg per mouse for 1 week as previously described³⁵.

Bulk RNA-seq analysis of murine $\gamma\delta$ T cells. To isolate $\gamma\delta$ T cells from *Kit*^{V558del/+} mice, CD45⁺CD3⁺NK1.1⁻TCR $\gamma\delta$ ⁺ T cells were sorted from vehicle and imatinib treated mice using a FACS Aria III (BD Biosciences). Total RNA was extracted, and next-generation RNA sequencing was performed on the Illumina HiSeq 2500 platform at the High-Throughput Sequencing Core of the Children's Hospital of Philadelphia. Sequencing reads were aligned to the mouse genome (mm10) using STAR aligner with default parameters⁴⁹. Expression was quantified using featureCounts⁵⁰. Differentially expressed genes (DEGs) were identified using the DESeq2 package v1.46.0 and fold changes were shrunk with apegglm. The relevant differential expression coefficient statistic was used as input for Gene Set Enrichment Analysis (GSEA) using the fgsea⁵¹ package. Pathways were sourced from the Molecular Signatures Database (MsigDB)⁵². For a comparison to our published murine scRNA-seq data, we utilized CIBERSORTx⁵³ using the scRNA-seq data as a reference with S-mode batch correction.

Statistical analysis. Statistical analyses for the scRNA-seq and TCR δ repertoires datasets are described above. Predicted 5-year recurrence free survival was calculated based on patient characteristics utilizing a published nomogram³¹. Overall survival analysis was performed using Kaplan-Meier method with the R package "survival." Significance was assessed with Logrank test,

and hazard ratio with a Cox proportional hazard model. Signature expression scores were calculated using a robust Z-scoring framework on normalized expression data from previously published datasets^{28,30}. Each gene was median-centered and scaled by a robust spread estimate combining the standard deviation and median absolute deviation (MAD) across samples. Standardized values were then averaged across signature genes to yield a per-sample score. Correlations between continuous variables were evaluated using Spearman's rank correlation coefficient. Differences between 2 experimental groups were analyzed using an unpaired, 2-tailed Welch's *t*-test or Mann-Whitney U test. All *in vitro* and *ex vivo* experiments were performed using at least three biological samples. Statistics analyses were carried out using Prism 9.0 (Graph Pad Software) and R software (R Foundation for Statistical Computing). Data are shown as mean $\pm$ SEM or median, as specified in the Fig. legends. A *p* value < 0.05 was considered statistically significant.

Data availability. scRNA-seq data from sorted human $\gamma\delta$ T cells and GIST tumors have been deposited to NCBI GEO under accession number GSE301351. Bulk RNA-seq data from sorted murine $\gamma\delta$ T cells are available under accession number GSE300283. TCR $\alpha\delta$ sequencing data from sorted human $\gamma\delta$ T cells are accessible via the ImmunoSEQ platform under the project name UPenn_00131343_R-01. Murine scRNA-seq used in this study were previously deposited in the Sequencing Read Archive (SRA) under accession number PRJNA859907. All remaining data supporting the findings of this study are available within the article and its supplementary materials or from the corresponding author upon reasonable request.

Code availability. Source code will be made available upon publication, and any code can be made available to the reviewers upon request.

Author contributions

S.Z. conceived and designed the study, performed all the experiments, and carried out formal analysis, validation, investigation, methodology. S.Z. also wrote the manuscript with input from all authors. J.H.S. and M.T.M performed computational and biological analyses of scRNA-seq data, including data interpretation, curation, methodology, formal analysis, and manuscript review and editing. Y.T.F. conducted all next-generation sequencing and analysis of TCR data, including interpretation and writing—review and editing. J.E.P. M.T.M and J.R. contributed to scRNA-seq biological analyses and data interpretation. K.T., H.K., T.H., K.D., F.R., M.T. M., S.J.L., and J.Q.Z. provided methodology and data analysis. All authors were involved in data curation, methodology, and manuscript review and editing. R.P.D. and X.X provided conceptualization and contributed to manuscript writing. R.P.D. provided supervision, funding acquisition, and project administration.

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Additional Information

Supplementary information and additional supporting data files are available for this paper. Correspondence and requests for materials should be addressed to Ronald P. DeMatteo (dematter@upenn.edu).

Figure Legends

Fig. 1: scRNA-seq analysis of $\gamma\delta$ T cells in PBMCs and tumors from patients with GIST.

T cells were sorted from PBMCs and tumor specimens and processed for 10x Genomics scRNA-seq. **A) left**, UMAP of $\gamma\delta$ T cells annotated by broad class (V δ 1, V δ 2, V δ 3) defined by canonical marker expression. **B)** UMAP of $\gamma\delta$ T cells annotated by cluster (55,228 cells across 4 PBMC and 2 tumor samples). **C) top**, Bar plots showing the distribution of each cell cluster across PBMC and tumor samples and **bottom**, across the 6 samples. **D)** Dot plot of selected differentially expressed genes in each cluster. Genes relevant to V δ 1, V δ 2, and V δ 3 subsets are indicated by color. Dot size indicates the percentage of cells in the cluster expressing the gene, and color indicates Z-score-normalized average expression between clusters. **E)** Differentially expressed genes between PBMCs and tumor samples shown as **left** a volcano plot and **right** gene set enrichment (GSEA) analysis of differentially expressed pathways. Representative significant pathways (FDR<0.1) are shown. No pathways were significantly upregulated in PBMC samples. **F) top**, Volcano plot of differentially expressed genes between V δ 1 and V δ 2 cells within tumor samples. **bottom**, GSEA analysis of differentially expressed pathways between tumor V δ 1 and V δ 2 cells. Representative significant pathways (FDR<0.1) are shown. **E-F)** Hallmark (H) and KEGG (K) pathways are indicated in parentheses. FDR, false discovery rate. NES, normalized enrichment score. **G)** CytoTRACE stemness score across $\gamma\delta$ T cells shown as **left** UMAP and **right** violin plots separated by V δ 1, V δ 2, and V δ 3 subsets in PBMCs and tumor samples.

Fig. 2: Regulon activity and potential gene-regulatory networks based on SCENIC of PBMC

and tumor $\gamma\delta$ T cells. **A)** Differential transcriptional regulons as inferred by SCENIC between PBMCs and tumor $\gamma\delta$ T cells. **B)** UMAP visualization of $\gamma\delta$ T cells colored by regulon activity scores calculated using AUCell across the dataset. **C)** Heatmap of selected regulon activity across $\gamma\delta$ T cells subsets. Regulon activity is Z-score normalized by row. **D)** Violin plots of selected regulon

activities across PBMC and tumor $\gamma\delta$ T cells stratified by broad class (V δ 1, V δ 2, V δ 3). Significance between tumor V δ 1 and tumor V δ 2 cells was assessed using the Wilcoxon rank-sum test. **E)** Gene module scores of broad $\gamma\delta$ T cell classes within PBMCs and tumors as in **D**. **F)** Enrichment analysis using Gene Ontology pathways was conducted on the predicted target genes of each regulon. Selected significant pathways are shown.

Fig. 3: Compositional and clonal analysis of $\gamma\delta$ T cells in PBMCs and tumors. $\gamma\delta$ T cells were sorted from human PBMC and GISTs. gDNA was processed for TCR $\alpha\delta$ CDR3 next-generation sequencing. **A)** Clonotype frequency in all specimens (n=11). **B)** Frequency of highly expanded (>10) clonotypes in 5 matched PBMC and tumors. **C)** Frequency of TCRDV gene segment usage in matched untreated (UT) and resistant (Res) specimens. **D)** Tree plots illustrating the composition of TCR repertoires in matched specimens. Each square represents a clone, with its area proportional to the clone size and its color indicating the V gene usage. **E)** Scatter plot of cell count distribution of a clone in PBMC versus tumor. Each point is a clone and the x,y axes denote the log transformed cell count in PBMC and tumor. Clones which have significantly high expansion are highlighted with FDR controlled q value. **F)** The alluvium plot for the VDJ combination of clones which have significantly high expansion in untreated and resistant tumors. **G)** The frequency change between the untreated and resistant patient groups for each VDJ recombination in clones which are highly expanded in tumor. VDJ recombinations with significant change are highlighted with “***”. **H)** Frequency of V δ 1 and V δ 2 $\gamma\delta$ T cells determined by flow cytometry in PBMCs and tumors. **I) left,** Frequency of V δ 1 and V δ 2 $\gamma\delta$ T cells in tumors or **right,** PBMCs based on treatment status in untreated (UT; n=15), and resistant (Res; n=10) specimens. Medians indicated. For all panels, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; **** $p < 0.0001$.

Fig. 4: Tumor-infiltrating V δ 1 T cells exhibit enhanced cytotoxicity and are prognostic of

619 **improved clinical outcomes. A)** Comparison of selected marker expression by flow cytometry
 620 between tumor-infiltrating V δ 1 and V δ 2 T cells. Data represent mean $\pm$ SEM. **B)** Summary of
 621 selected markers in PBMC-derived (n=10) and tumor-derived (n=8-16) V δ 1 T cells. Data
 622 represent mean $\pm$ SEM. **C)** Tumor $\gamma\delta$ T cells were sorted and grouped by relatively high V δ 1 (mean
 623 59.2%, n=6) or low V δ 1 frequency (mean 32.6%, n=3) using a cut-off of 50% V δ 1 frequency. **D)**
 624 qPCR showing mRNA expression of *TRGC2*, *TRDV3*, *IFNG* and *KLRK1* in sorted tumor $\gamma\delta$ T cells
 625 with high or low V δ 1 proportions. Data represent mean $\pm$ SEM. **E)** Composition of $\gamma\delta$ T cell subsets
 626 (n = 6,970 cells) in scRNA-seq data from untreated (n=3) and resistant (n=2) GISTs based on
 627 sorted $\gamma\delta$ T cell reference (Fig. 1A). **F) left**, Volcano plot of differentially expressed genes between
 628 $\gamma\delta$ T cells in untreated (n=3) and resistant (n=2) GISTs based on scRNA-seq data. Key V δ 1 and
 629 V δ 2 marker genes (Fig. 1F) are highlighted. **right**, Corresponding Dot plot of V δ 1 and V δ 2 marker
 630 genes. **G)** Over representation analysis using Gene Ontology (GO), KEGG (K), and Reactome
 631 (R) gene sets in the top differentially expressed genes between untreated (**top**) and resistant $\gamma\delta$
 632 T cells (**bottom**). Representative significant pathways are shown (FDR <0.05). **H)** qPCR analysis
 633 showing relative mRNA expression of *TRGC2*, *TRDV3* and *IFNG* in untreated (n=15) and
 634 resistant (n=14) GISTs. Medians indicated. **I)** Flow cytometry analysis of CD107a expression on
 635 tumor V δ 1 $\gamma\delta$ T cells from untreated (UT; n=8) and resistant GISTs (Res; n=5). Data represent
 636 mean $\pm$ SEM. **J)** Summary data of effector status of tumor V δ 1 T cells from untreated (UT; n=9)
 637 and imatinib-resistant tumors based on flow cytometry (Res; n=3). Medians indicated. T_{EM} -
 638 effector memory T cells, T_{EMRA} - terminally differentiated effector memory T cells, T_{CM} - central
 639 memory T cells, T_N - naïve T cells. **K)** Kaplan-Meier overall survival analysis of a published
 640 Japanese cohort of 32 patients with untreated primary GISTs, stratified by median expression
 641 score of $\gamma\delta$ T cell signature genes (*TRGC2*, *TRDC*, *TRDV3*, and *TRGV2*). **L)** Scatterplot of
 642 predicted 5-year recurrence-free survival (RFS) based on median expression score of $\gamma\delta$ T cell
 643 signature genes in our published cohort of bulk RNA-seq in 30 untreated primary GISTs. **M)**

Scatterplot of predicted 5-year RFS in 17 patients with untreated primary GIST based on V δ 1 percentage of $\gamma\delta$ T cells determined by flow cytometry. For all panels, * p <0.05; ** p <0.01; *** p <0.001, **** p <0.0001.

Fig. 5: Expanded circulating and tumor-infiltrating $\gamma\delta$ T cells exhibit cytotoxicity against GIST. **A)** Expanded $\gamma\delta$ T cells from healthy donors were co-cultured with luciferase-expressing GIST-T1 cells at the indicated effector-target (E:T) ratios. After 24h, cytotoxicity was assessed by measuring luciferase activity. **B)** Representative images of a tumor spheroid assay. GIST-T1 cell spheroids were co-cultured with expanded $\gamma\delta$ T cells, E:T ratio of 2.5:1, for 24h. Tumor cells expressed luciferase (green), and $\gamma\delta$ T cells were labeled with CellTrace (red). **C)** Cytotoxicity in spheroids after co-culture with $\gamma\delta$ T cells for 72h at the indicated E:T ratios as measured by luciferase assay. **D) left,** Histograms of KIT expression on tumor cells and **right,** quantification of KIT⁺ tumor cell killing in human GIST tissue slices co-cultured with expanded $\gamma\delta$ T cells at an E:T ratio of 5:1 for 24h, assessed by flow cytometry after tissue slice dissociation. **E)** Representative pictures showing expansion of tumor-derived $\gamma\delta$ T cells on day 6. Tumor-infiltrating lymphocytes (TILs) from untreated GISTs were isolated using CD45 microbeads and expanded with anti- $\gamma\delta$ TCR antibody supplemented with IL-2 (200IU/ml). **F)** Representative flow plots showing robust expansion of V δ 1 cells at 24h following co-culture of expanded tumor-derived $\gamma\delta$ T cells with tumor single-cell suspensions (tumor SC). **G)** Cytotoxicity against autologous KIT⁺ tumor cells was evaluated by co-culturing expanded tumor-derived $\gamma\delta$ T cells as shown in **F. left,** Flow cytometry histograms of KIT expression on tumor cells and **right,** quantification of KIT⁺ tumor cell killing. **H) left,** Flow cytometry histograms of MICA expression on tumor cells (KIT⁺) and T cells (CD45⁺CD3⁺) and **right,** qPCR showing relative *MICA* mRNA expression in KIT⁺ tumor cells and KIT⁻ cells from GIST patients. **I)** Tumor-infiltrating lymphocytes (TILs) were treated with recombinant MICA (rhMICA, 10 μ g/ml) and IL-2 for 72h and analyzed by flow cytometry,

demonstrating an increased percentage of V δ 1 $\gamma\delta$ T cells, and greater frequency of HLA-DR among the same. **J)** Sorted tumor $\gamma\delta$ T cells from untreated GISTs were co-cultured with rhMICA or anti-NKG2D antibody for 24h, and IFN γ production was measured by ELISpot. Data represent mean $\pm$ SEM, For all panels, * p <0.05; ** p <0.01; *** p <0.001, **** p <0.0001.

Fig. 6: Imatinib drives functional and compositional reprogramming of tumor $\gamma\delta$ T cells *in vivo*. $\gamma\delta$ T cells were sorted from *Kit*^{V558del/+} mice treated with vehicle or imatinib for 1 week and processed for bulk RNA-seq. **A)** Volcano plot of differentially expressed genes in tumor $\gamma\delta$ T cells based on treatment (n=3/group). Significant genes defined with p-adjusted < 0.05 and log₂FC > 1. **B)** Gene set enrichment analysis (GSEA) of Hallmark (H) and KEGG (K) pathways in tumor $\gamma\delta$ T cells based on treatment. **C)** Selected enrichment plots from **B**. NES, normalized enrichment score; FDR, false discovery rate. **D)** Heatmap of selected tumor $\gamma\delta$ T cell signature genes across samples. Color represents Z-score normalized expression for each gene. **E)** UMAP of previously published scRNA-seq data of murine $\gamma\delta$ T cells (1,641) cells, separated by treatment and colored by *Tcr γ 6* gene expression. **F)** Deconvolution of bulk RNA-seq data with the single-cell reference in **E** using CIBERSORTx. Significance was assessed using a Welch's 2-sided t-test. **G)** qPCR of expression of selected genes from sorted $\gamma\delta$ T cells. n=4-6/group. **H)** Tumor weight after *Kit*^{V558del/+} mice were treated for 1 week with vehicle, imatinib, or imatinib plus anti-PDL1. n=9/group. **I)** The frequency of V γ 7 and V γ 6 high subsets among tumor $\gamma\delta$ T cells from *Kit*^{V558del/+} mice treated in **H** as assessed by flow cytometry. **J)** The frequency of PD1⁺ cells in V γ 7 and V γ 6 high subsets among tumor $\gamma\delta$ T cells in each group. **K-L)** The frequency of IFN γ ⁺ cells (**K**) and IL17A⁺ cells (**L**) among tumor $\gamma\delta$ T cells from *Kit*^{V558del/+} mice treated in **H**. Data represent mean $\pm$ SEM. n=4/group. For all panels, * p <0.05, ** p <0.01, *** p <0.001, **** p <0.0001.

References

1. Blay, J.Y., Kang, Y.K., Nishida, T. & von Mehren, M. Gastrointestinal stromal tumours. *Nat Rev Dis Primers* **7**, 22 (2021).
2. Balachandran, V.P. *et al.* Imatinib potentiates antitumor T cell responses in gastrointestinal stromal tumor through the inhibition of Ido. *Nat Med* **17**, 1094–100 (2011).
3. Sommer, G. *et al.* Gastrointestinal stromal tumors in a mouse model by targeted mutation of the Kit receptor tyrosine kinase. *Proc Natl Acad Sci U S A* **100**, 6706–11 (2003).
4. Etherington, M.S. *et al.* Tyrosine Kinase Inhibition Activates Intratumoral gammadelta T Cells in Gastrointestinal Stromal Tumor. *Cancer Immunol Res* **12**, 107–119 (2024).
5. Singh, A.S. *et al.* A Randomized Phase II Study of Nivolumab Monotherapy or Nivolumab Combined with Ipilimumab in Patients with Advanced Gastrointestinal Stromal Tumors. *Clin Cancer Res* **28**, 84–94 (2022).
6. Hayday, A., Dechanet-Merville, J., Rossjohn, J. & Silva-Santos, B. Cancer immunotherapy by gammadelta T cells. *Science* **386**, eabq7248 (2024).
7. Silva-Santos, B., Mensurado, S. & Coffelt, S.B. gammadelta T cells: pleiotropic immune effectors with therapeutic potential in cancer. *Nat Rev Cancer* **19**, 392–404 (2019).
8. Wu, Y. *et al.* A local human Vdelta1 T cell population is associated with survival in nonsmall-cell lung cancer. *Nat Cancer* **3**, 696–709 (2022).
9. Gentles, A.J. *et al.* The prognostic landscape of genes and infiltrating immune cells across human cancers. *Nat Med* **21**, 938–945 (2015).
10. Stary, V. *et al.* Dysfunctional tumor-infiltrating Vdelta1 + T lymphocytes in microsatellite-stable colorectal cancer. *Nat Commun* **15**, 6949 (2024).
11. Wu, P. *et al.* gammadeltaT17 cells promote the accumulation and expansion of myeloid-derived suppressor cells in human colorectal cancer. *Immunity* **40**, 785–800 (2014).
12. Petroni, G. *et al.* IL-17A-secreting gammadelta T cells promote resistance to CDK4/CDK6 inhibitors in HR(+)HER2(-) breast cancer via CX3CR1(+) macrophages. *Nat Cancer* **6**, 1656–1675 (2025).
13. Reis, B.S. *et al.* TCR-Vgammadelta usage distinguishes protumor from antitumor intestinal gammadelta T cell subsets. *Science* **377**, 276–284 (2022).
14. Melandri, D. *et al.* The gammadeltaTCR combines innate immunity with adaptive immunity by utilizing spatially distinct regions for agonist selection and antigen responsiveness. *Nat Immunol* **19**, 1352–1365 (2018).
15. Rancan, C. *et al.* Exhausted intratumoral Vdelta2(-) gammadelta T cells in human kidney cancer retain effector function. *Nat Immunol* **24**, 612–624 (2023).
16. Davies, D. *et al.* PD-1 defines a distinct, functional, tissue-adapted state in Vdelta1(+) T cells with implications for cancer immunotherapy. *Nat Cancer* **5**, 420–432 (2024).
17. Pizzolato, G. *et al.* Single-cell RNA sequencing unveils the shared and the distinct cytotoxic hallmarks of human TCRVdelta1 and TCRVdelta2 gammadelta T lymphocytes. *Proc Natl Acad Sci U S A* **116**, 11906–11915 (2019).
18. Payne, K.K. *et al.* BTN3A1 governs antitumor responses by coordinating alphabeta and gammadelta T cells. *Science* **369**, 942–949 (2020).
19. Tan, L. *et al.* A fetal wave of human type 3 effector gammadelta cells with restricted TCR diversity persists into adulthood. *Sci Immunol* **6**(2021).

- 738 20. Gulati, G.S. *et al.* Single-cell transcriptional diversity is a hallmark of developmental
739 potential. *Science* **367**, 405–411 (2020).
- 740 21. Bravo Gonzalez-Blas, C. *et al.* SCENIC+: single-cell multiomic inference of enhancers
741 and gene regulatory networks. *Nat Methods* **20**, 1355–1367 (2023).
- 742 22. McMurray, J.L. *et al.* Transcriptional profiling of human Vdelta1 T cells reveals a
743 pathogen-driven adaptive differentiation program. *Cell Rep* **39**, 110858 (2022).
- 744 23. Tille, L. *et al.* Activation of the transcription factor NFAT5 in the tumor
745 microenvironment enforces CD8(+) T cell exhaustion. *Nat Immunol* **24**, 1645–1653
746 (2023).
- 747 24. Li, R. *et al.* Analysis of the three-dimensional genome of exhausted CD8(+) T cells
748 reveals a critical role of IRF8 in their differentiation and functions in cancer. *Nat*
749 *Immunol* **26**, 2280–2295 (2025).
- 750 25. Kurd, N.S. *et al.* Early precursors and molecular determinants of tissue-resident memory
751 CD8(+) T lymphocytes revealed by single-cell RNA sequencing. *Sci Immunol* **5**(2020).
- 752 26. Kumar, B.V. *et al.* Human Tissue-Resident Memory T Cells Are Defined by Core
753 Transcriptional and Functional Signatures in Lymphoid and Mucosal Sites. *Cell Rep* **20**,
754 2921–2934 (2017).
- 755 27. Godec, J. *et al.* Compendium of Immune Signatures Identifies Conserved and Species-
756 Specific Biology in Response to Inflammation. *Immunity* **44**, 194–206 (2016).
- 757 28. Yamaguchi, U. *et al.* Distinct gene expression-defined classes of gastrointestinal stromal
758 tumor. *J Clin Oncol* **26**, 4100–8 (2008).
- 759 29. Yu, X. *et al.* Pan-cancer gammadelta TCR analysis uncovers clonotype diversity and
760 prognostic potential. *Cell Rep Med* **5**, 101764 (2024).
- 761 30. Vitiello, G.A. *et al.* Differential immune profiles distinguish the mutational subtypes of
762 gastrointestinal stromal tumor. *J Clin Invest* **129**, 1863–1877 (2019).
- 763 31. Gold, J.S. *et al.* Development and validation of a prognostic nomogram for recurrence-
764 free survival after complete surgical resection of localised primary gastrointestinal
765 stromal tumour: a retrospective analysis. *Lancet Oncol* **10**, 1045–52 (2009).
- 766 32. Zhou, J., Kang, N., Cui, L., Ba, D. & He, W. Anti-gammadelta TCR antibody-expanded
767 gammadelta T cells: a better choice for the adoptive immunotherapy of lymphoid
768 malignancies. *Cell Mol Immunol* **9**, 34–44 (2012).
- 769 33. Lopes, N. *et al.* Distinct metabolic programs established in the thymus control effector
770 functions of gammadelta T cell subsets in tumor microenvironments. *Nat Immunol* **22**,
771 179–192 (2021).
- 772 34. Lien, S.C. *et al.* Tumor reactive gammadelta T cells contribute to a complete response to
773 PD-1 blockade in a Merkel cell carcinoma patient. *Nat Commun* **15**, 1094 (2024).
- 774 35. Seifert, A.M. *et al.* PD-1/PD-L1 Blockade Enhances T-cell Activity and Antitumor
775 Efficacy of Imatinib in Gastrointestinal Stromal Tumors. *Clin Cancer Res* **23**, 454–465
776 (2017).
- 777 36. Ou, L. *et al.* Dichotomous and stable gamma delta T-cell number and function in healthy
778 individuals. *J Immunother Cancer* **9**(2021).
- 779 37. Harly, C. *et al.* Human gammadelta T cell sensing of AMPK-dependent metabolic tumor
780 reprogramming through TCR recognition of EphA2. *Sci Immunol* **6**(2021).
- 781 38. Groh, V. *et al.* Broad tumor-associated expression and recognition by tumor-derived
782 gamma delta T cells of MICA and MICB. *Proc Natl Acad Sci U S A* **96**, 6879–84 (1999).

39. Yuan, L. *et al.* Phosphoantigens glue butyrophilin 3A1 and 2A1 to activate Vgamma9Vdelta2 T cells. *Nature* **621**, 840–848 (2023).
40. Cavnar, M.J. *et al.* KIT oncogene inhibition drives intratumoral macrophage M2 polarization. *J Exp Med* **210**, 2873–86 (2013).
41. Medina, B.D. *et al.* Oncogenic kinase inhibition limits Batf3-dependent dendritic cell development and antitumor immunity. *J Exp Med* **216**, 1359–1376 (2019).
42. Ferry, G.M. *et al.* A Simple and Robust Single-Step Method for CAR-Vdelta1 gammadeltaT Cell Expansion and Transduction for Cancer Immunotherapy. *Front Immunol* **13**, 863155 (2022).
43. Makkouk, A. *et al.* Off-the-shelf Vdelta1 gamma delta T cells engineered with glypican-3 (GPC-3)-specific chimeric antigen receptor (CAR) and soluble IL-15 display robust antitumor efficacy against hepatocellular carcinoma. *J Immunother Cancer* **9**(2021).
44. Sterner, R.C. & Sterner, R.M. CAR-T cell therapy: current limitations and potential strategies. *Blood Cancer J* **11**, 69 (2021).
45. Hao, Y. *et al.* Dictionary learning for integrative, multimodal and scalable single-cell analysis. *Nat Biotechnol* **42**, 293–304 (2024).
46. Kanehisa, M., Furumichi, M., Tanabe, M., Sato, Y. & Morishima, K. KEGG: new perspectives on genomes, pathways, diseases and drugs. *Nucleic Acids Res* **45**, D353–D361 (2017).
47. Aibar, S. *et al.* SCENIC: single-cell regulatory network inference and clustering. *Nat Methods* **14**, 1083–1086 (2017).
48. Van de Sande, B. *et al.* A scalable SCENIC workflow for single-cell gene regulatory network analysis. *Nat Protoc* **15**, 2247–2276 (2020).
49. Dobin, A. *et al.* STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* **29**, 15–21 (2013).
50. Liao, Y., Smyth, G.K. & Shi, W. featureCounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* **30**, 923–30 (2014).
51. Korotkevich, G. *et al.* Fast gene set enrichment analysis. *bioRxiv*, 060012 (2021).
52. Liberzon, A. *et al.* The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst* **1**, 417–425 (2015).
53. Newman, A.M. *et al.* Determining cell type abundance and expression from bulk tissues with digital cytometry. *Nat Biotechnol* **37**, 773–782 (2019).

Figure 2

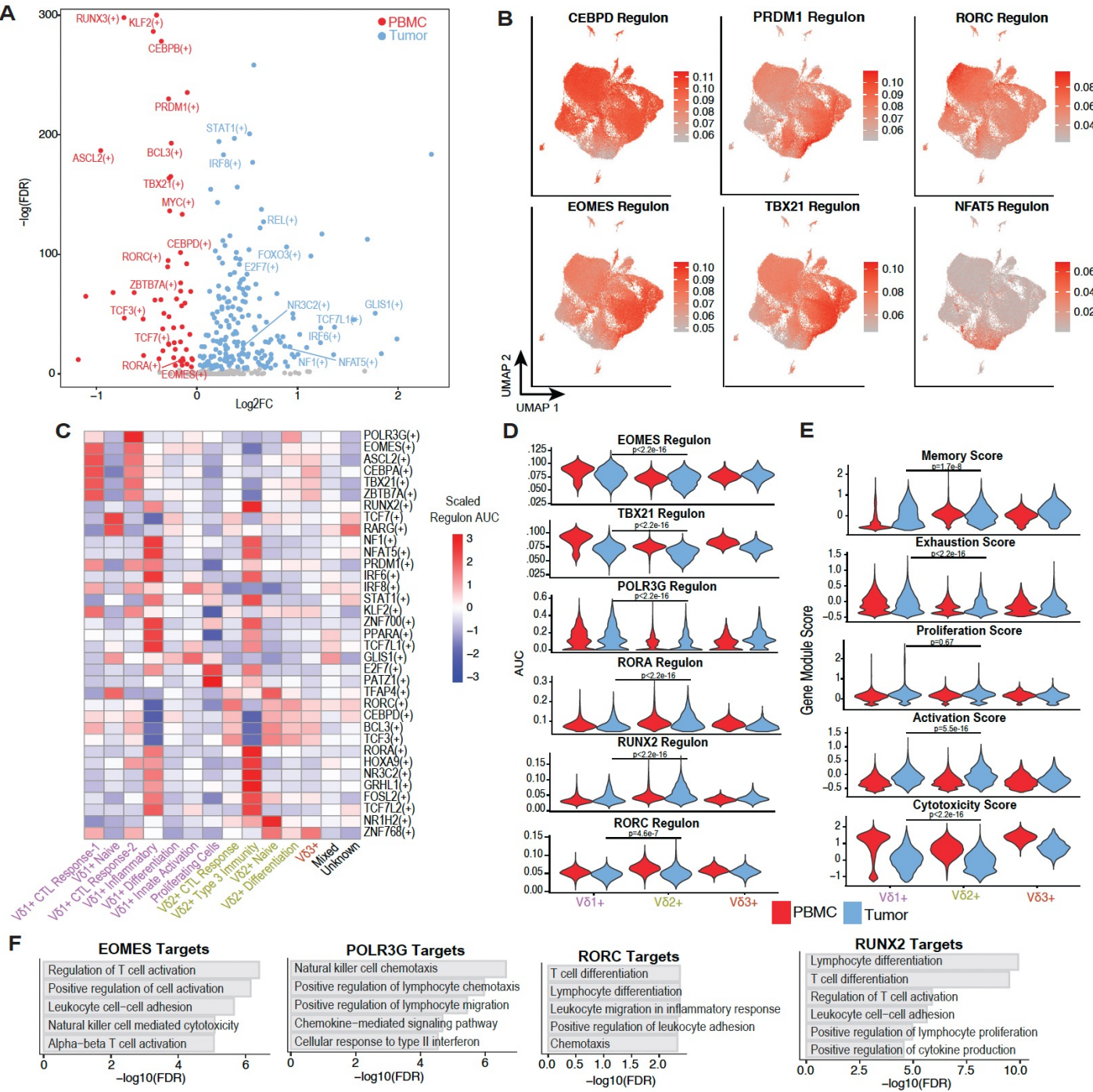


Figure 3

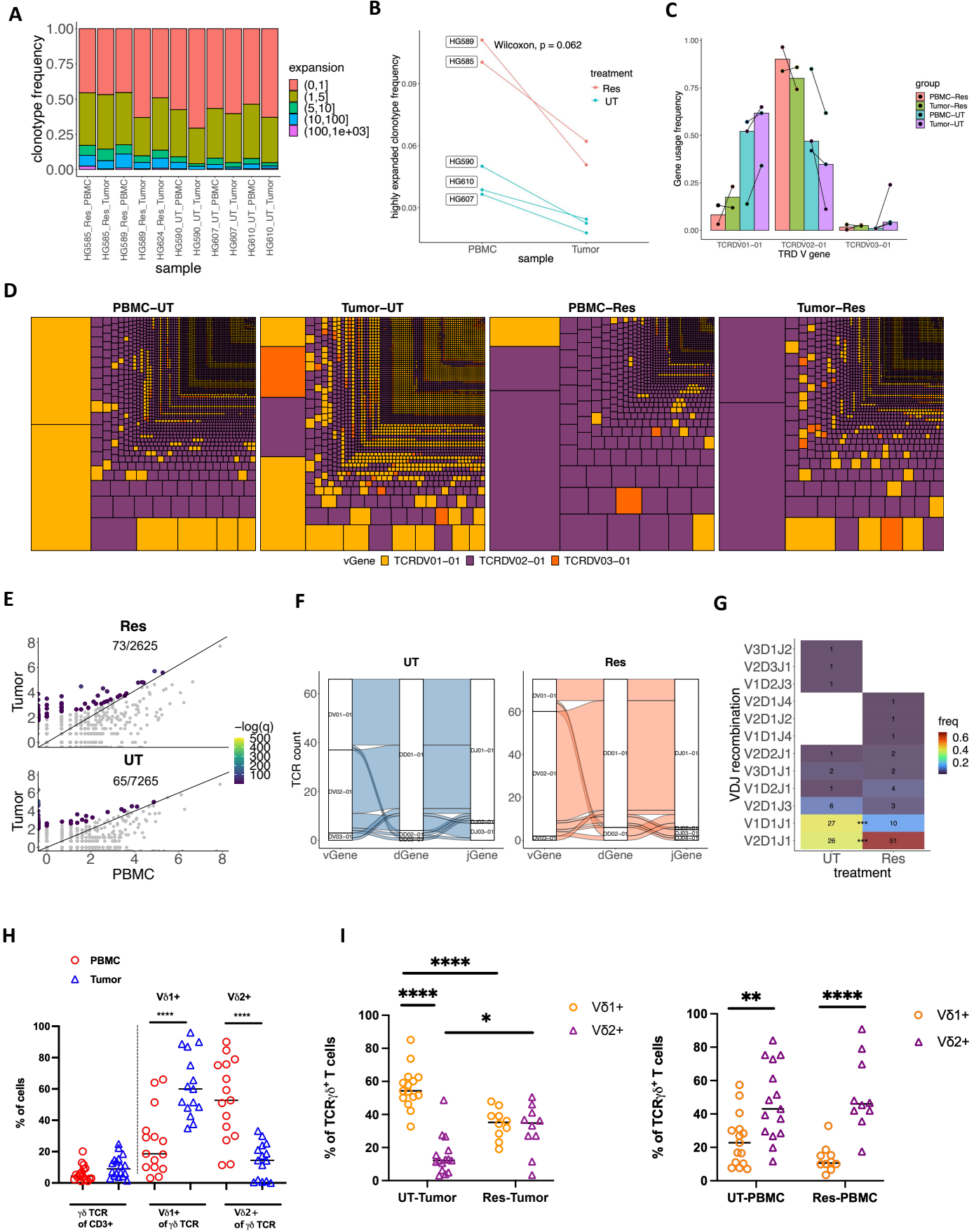


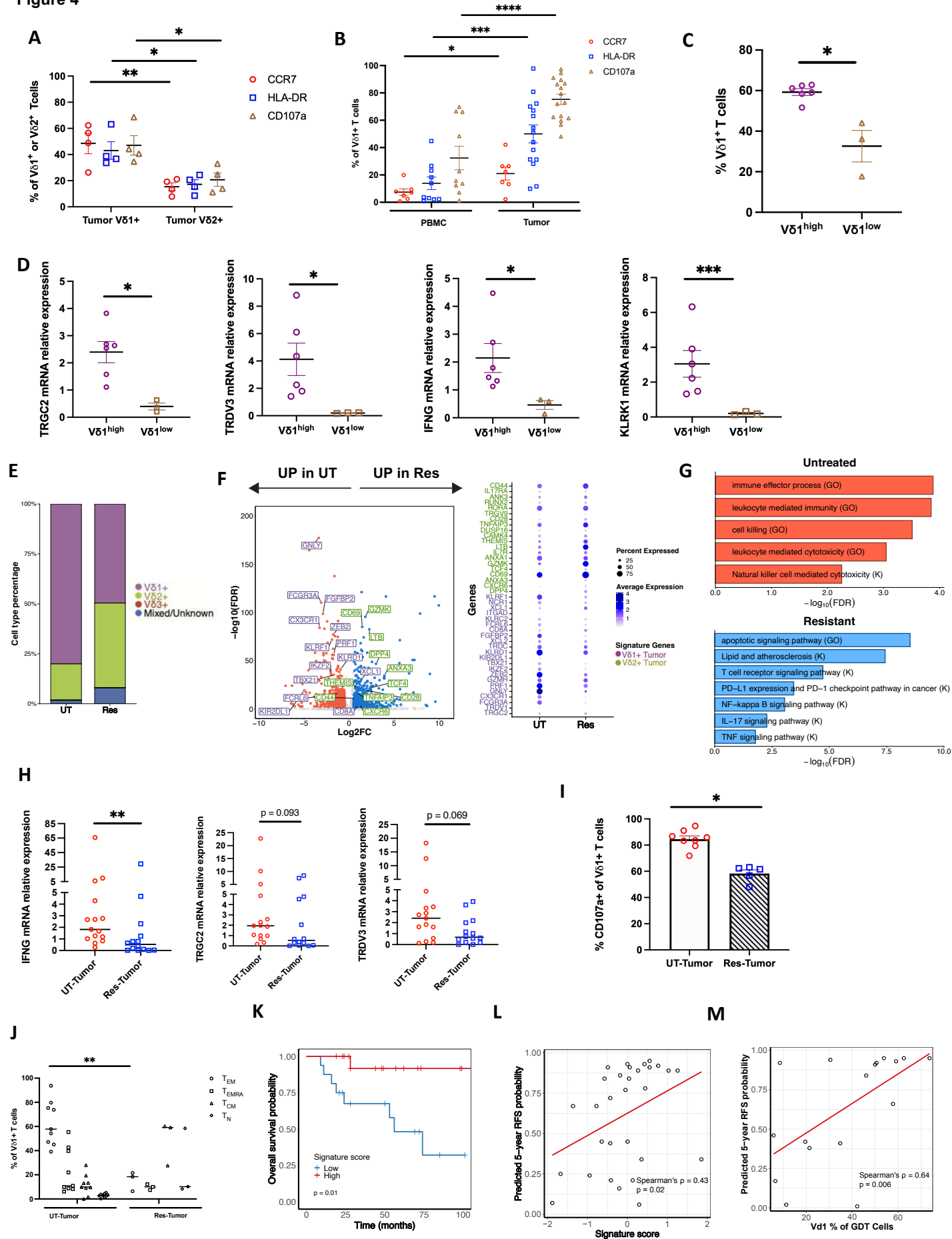
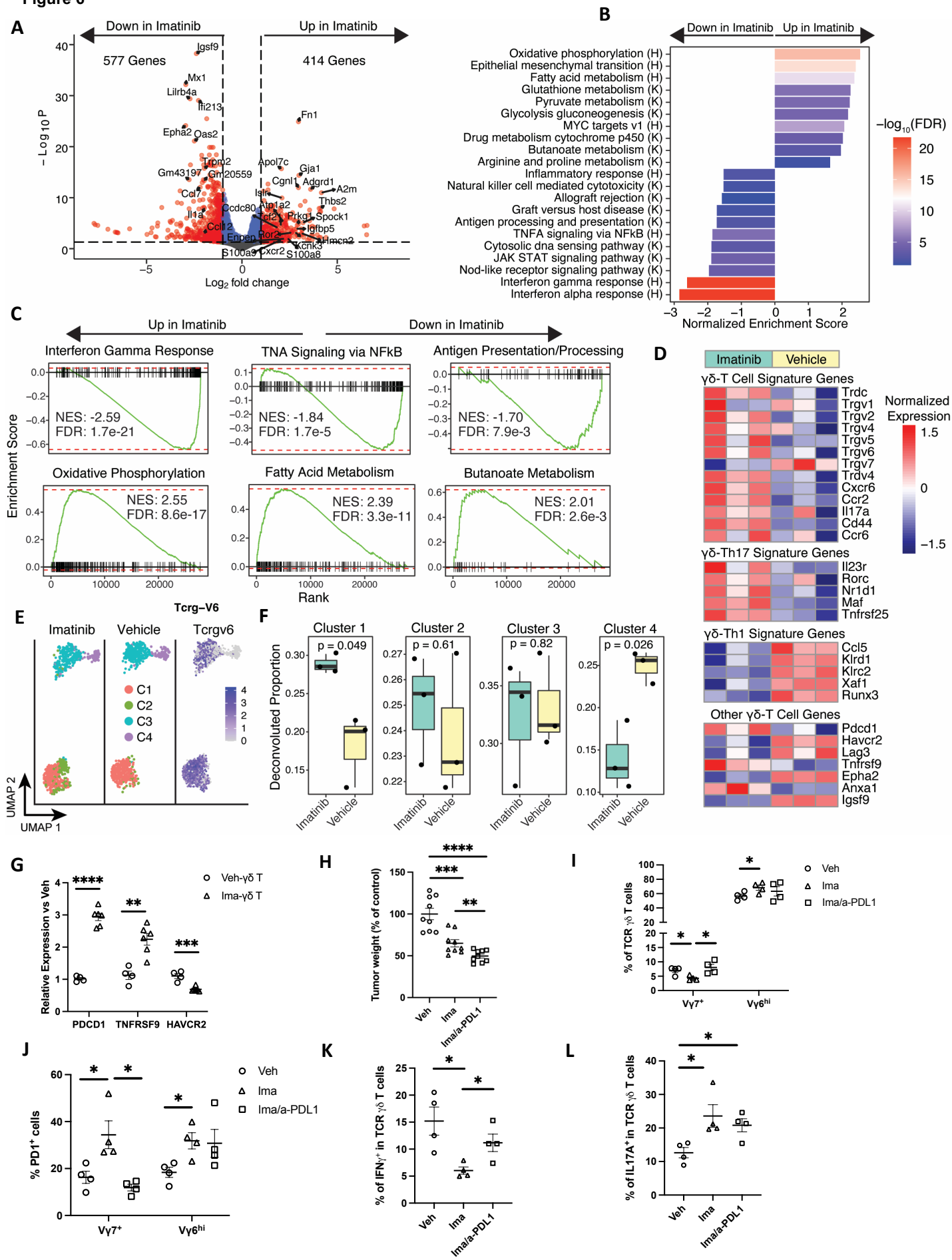
Figure 4

Figure 6



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